

Intestinal microbiota metabolism of L-carnitine, a nutrient in red meat, promotes atherosclerosis



Intestinal microbiota metabolism of choline/phosphatidylcholine produces trimethylamine (TMA), which is further metabolized to a proatherogenic species, trimethylamine-N-oxide (TMAO). Herein we demonstrate that intestinal microbiota metabolism of dietary L-carnitine, a trimethylamine abundant in red meat, also produces TMAO and accelerates atherosclerosis. Omnivorous subjects are shown to produce significantly more TMAO than vegans/vegetarians following ingestion of L-carnitine through a microbiota-dependent mechanism. Specific bacterial taxa in human feces are shown to associate with both plasma TMAO and dietary status. Plasma L-carnitine levels in subjects undergoing cardiac evaluation (n = 2,595) predict increased risks for both prevalent cardiovascular disease (CVD) and incident major adverse cardiac events (MI, stroke or death), but only among subjects with concurrently high TMAO levels. Chronic dietary L-carnitine supplementation in mice significantly altered cecal microbial composition, markedly enhanced synthesis of TMA/TMAO, and increased atherosclerosis, but not following suppression of intestinal microbiota. Dietary supplementation of TMAO, or either carnitine or choline in mice with intact intestinal microbiota, significantly reduced reverse cholesterol transport in vivo. Intestinal microbiota may thus participate in the well-established link between increased red meat consumption and CVD risk.